

SNA Fall 2025: Assignment 2

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Assignment 2 Description

Leave the link to the Assignment description but delete these instructions before submitting:

- Please keep the overall structure of this `.Rmd` file (Task 1, Task 2, Task 3 – including subtasks and instructions), but feel free to customize other elements, such as the YAML header, heading hierarchy, or rephrase titles, to suit your style.
- Knit the file as a PDF or Word document only, and omit lengthy outputs. Remember that `.Rmd` files will not knit if there are any errors, even small typos: test knitting before the deadline

Libraries and Data

```
# install.packages("kableExtra")
```

```
library(igraph)
library(statnet)
library(intergraph)
library(tidyverse)
library(ggraph)
library(kableExtra)
load("luke.datasets.RData")
```

```
describe.ICTS_G10()
```

```
class(ICTS_G10)
```

```
## [1] "network"
```

```
network::list.vertex.attributes(ICTS_G10)
```

```
## [1] "DiscCol"      "DiscExp"      "vertex.names" "year"
```

```
network::list.edge.attributes(ICTS_G10)
```

```
## [1] "na"
```

```
ICTS_G10_matrix <- as.matrix(ICTS_G10)
vn <- ICTS_G10 %v% "vertex.names"
dimnames(ICTS_G10_matrix) <- list(vn, vn)
knitr::kable(ICTS_G10_matrix[1:15, 1:15])
```

	380	383	385	387	388	390	391	392	393	394	396	397	404	407	410
380	0	1	1	0	1	0	0	0	0	1	1	0	1	0	0
383	1	0	0	1	0	0	0	0	0	1	0	0	0	0	0
385	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
387	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0
388	1	0	0	0	0	0	0	0	0	1	0	0	1	0	1
390	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0
391	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
392	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
393	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1
394	1	1	0	0	1	0	0	0	0	0	0	0	1	0	0
396	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
397	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
404	1	0	0	0	1	1	0	0	0	1	0	0	0	0	0
407	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
410	0	0	0	0	1	0	0	0	1	0	0	0	0	0	0

```
write.csv(ICTS_G10_matrix, file = "ICTS_G10.csv")
```

```
ICTS_G10_df <- read.csv("ICTS_G10.csv",
  header = TRUE,
  row.names = 1,
  as.is = TRUE)
```

```
class(ICTS_G10_df)
```

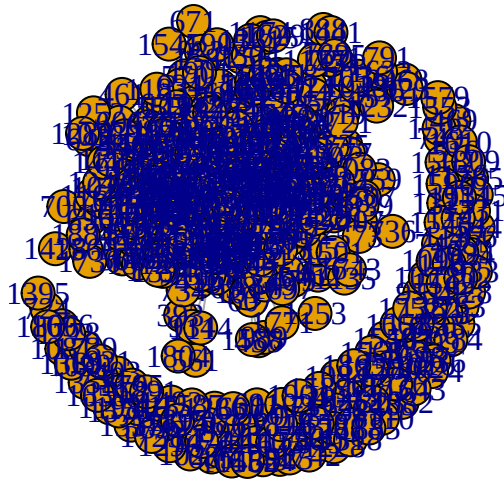
```
## [1] "data.frame"
```

Explore the Network

```
nICTS_G10 <- igraph::graph_from_adjacency_matrix(ICTS_G10_matrix, mode = "undirected")
class(nICTS_G10)
```

```
## [1] "igraph"
```

```
set.seed(42)
plot(nICTS_G10, displaylabels = TRUE, boxed.labels = FALSE, label.cex = 0.75)
```



```
gorder(nICTS_G10)
```

```
## [1] 493
```

```
gsize(nICTS_G10)
```

```
## [1] 1359
```

```
graph.density(nICTS_G10)
```

```
## [1] 0.01120566
```

```
diameter(nICTS_G10)
```

```
## [1] 9
```

```
edge_attr(nICTS_G10)
```

```
## named list()
```

```
vertex_attr(nICTS_G10)
```

```
## $name
## [1] "380" "383" "385" "387" "388" "390" "391" "392" "393" "394"
## [11] "396" "397" "404" "407" "410" "411" "414" "415" "419" "435"
## [21] "436" "449" "451" "452" "454" "455" "457" "458" "461" "462"
## [31] "464" "465" "466" "468" "469" "470" "471" "472" "475" "476"
## [41] "478" "479" "482" "483" "487" "488" "493" "496" "501" "502"
## [51] "504" "505" "506" "507" "513" "516" "525" "526" "531" "535"
## [61] "537" "538" "539" "543" "544" "545" "546" "547" "548" "549"
## [71] "550" "551" "552" "554" "555" "558" "560" "561" "565" "566"
## [81] "568" "570" "571" "573" "574" "576" "587" "589" "592" "597"
## [91] "599" "601" "602" "604" "606" "607" "612" "614" "616" "617"
## [101] "618" "619" "621" "622" "626" "630" "633" "634" "635" "638"
## [111] "639" "640" "642" "643" "644" "645" "646" "647" "649" "652"
## [121] "653" "654" "656" "666" "668" "669" "670" "671" "672" "673"
## [131] "675" "678" "685" "686" "687" "688" "691" "695" "696" "699"
## [141] "700" "701" "703" "704" "705" "706" "707" "708" "709" "715"
## [151] "716" "717" "718" "719" "720" "724" "725" "726" "729" "730"
## [161] "733" "734" "735" "736" "737" "738" "742" "745" "747" "749"
## [171] "751" "752" "758" "759" "760" "763" "765" "767" "769" "770"
## [181] "772" "776" "777" "790" "791" "878" "885" "886" "888" "894"
```

```

## [191] "900" "911" "912" "922" "923" "924" "925" "927" "928" "929"
## [201] "931" "932" "933" "934" "935" "936" "942" "943" "945" "946"
## [211] "947" "950" "951" "953" "960" "968" "977" "989" "992" "1001"
## [221] "1009" "1010" "1012" "1016" "1017" "1018" "1020" "1025" "1026" "1027"
## [231] "1030" "1032" "1034" "1043" "1046" "1047" "1054" "1059" "1067" "1068"
## [241] "1073" "1077" "1078" "1079" "1081" "1082" "1083" "1089" "1095" "1096"
## [251] "1099" "1102" "1103" "1108" "1110" "1111" "1113" "1114" "1129" "1134"
## [261] "1135" "1137" "1150" "1154" "1155" "1163" "1164" "1165" "1167" "1168"
## [271] "1171" "1173" "1174" "1175" "1176" "1177" "1181" "1182" "1183" "1187"
## [281] "1189" "1191" "1198" "1202" "1205" "1210" "1213" "1214" "1215" "1216"
## [291] "1217" "1228" "1232" "1233" "1239" "1241" "1244" "1251" "1252" "1253"
## [301] "1256" "1263" "1276" "1281" "1282" "1283" "1284" "1285" "1286" "1288"
## [311] "1295" "1296" "1301" "1302" "1304" "1305" "1306" "1312" "1318" "1325"
## [321] "1330" "1334" "1339" "1340" "1345" "1347" "1348" "1349" "1353" "1355"
## [331] "1357" "1358" "1359" "1361" "1365" "1366" "1369" "1370" "1371" "1372"
## [341] "1375" "1376" "1377" "1379" "1382" "1388" "1390" "1394" "1395" "1404"
## [351] "1409" "1414" "1428" "1429" "1432" "1440" "1441" "1446" "1448" "1450"
## [361] "1455" "1467" "1475" "1492" "1501" "1507" "1510" "1516" "1518" "1522"
## [371] "1523" "1526" "1529" "1532" "1536" "1539" "1540" "1541" "1542" "1545"
## [381] "1548" "1552" "1553" "1554" "1555" "1556" "1558" "1559" "1560" "1561"
## [391] "1562" "1567" "1568" "1575" "1576" "1579" "1580" "1581" "1582" "1583"
## [401] "1585" "1587" "1588" "1589" "1590" "1592" "1600" "1604" "1610" "1614"
## [411] "1622" "1623" "1624" "1625" "1628" "1629" "1631" "1632" "1637" "1642"
## [421] "1643" "1645" "1648" "1649" "1650" "1651" "1656" "1658" "1659" "1664"
## [431] "1668" "1675" "1696" "1697" "1698" "1699" "1721" "1722" "1723" "1730"
## [441] "1731" "1736" "1738" "1739" "1741" "1742" "1743" "1744" "1752" "1757"
## [451] "1771" "1773" "1776" "1778" "1783" "1784" "1791" "1795" "1802" "1804"
## [461] "1806" "1809" "1813" "1815" "1818" "1820" "1821" "1823" "1827" "1834"
## [471] "1841" "1842" "1843" "1864" "1865" "1872" "1873" "1879" "1883" "1884"
## [481] "1885" "1889" "1891" "1895" "1899" "1901" "1903" "1904" "1907" "1911"
## [491] "1913" "1920" "2314"

```

V(nICTS_G10)

```

## + 493/493 vertices, named, from 4835b5a:
## [1] 380 383 385 387 388 390 391 392 393 394 396 397 404 407 410
## [16] 411 414 415 419 435 436 449 451 452 454 455 457 458 461 462
## [31] 464 465 466 468 469 470 471 472 475 476 478 479 482 483 487
## [46] 488 493 496 501 502 504 505 506 507 513 516 525 526 531 535
## [61] 537 538 539 543 544 545 546 547 548 549 550 551 552 554 555
## [76] 558 560 561 565 566 568 570 571 573 574 576 587 589 592 597
## [91] 599 601 602 604 606 607 612 614 616 617 618 619 621 622 626
## [106] 630 633 634 635 638 639 640 642 643 644 645 646 647 649 652
## [121] 653 654 656 666 668 669 670 671 672 673 675 678 685 686 687
## [136] 688 691 695 696 699 700 701 703 704 705 706 707 708 709 715
## + ... omitted several vertices

```

E(nICTS_G10)

```

## + 1359/1359 edges from 4835b5a (vertex names):
## [1] 380--383 380--385 380--388 380--394 380--396 380--404 380--464
## [8] 380--470 380--472 380--478 380--501 380--549 380--550 380--571
## [15] 380--618 380--645 380--656 380--669 380--726 380--885 380--936
## [22] 380--992 380--1059 380--1067 380--1134 380--1181 380--1210 380--1339
## [29] 380--1347 380--1359 380--1370 380--1379 380--1390 380--1448 380--1592

```

```
## [36] 380--1783 380--1813 380--1864 380--1872 383--387 383--394 383--435
## [43] 383--458 383--464 383--470 383--472 383--478 383--507 383--601
## [50] 383--626 383--630 383--643 383--645 383--656 383--738 383--912
## [57] 383--947 383--1009 383--1012 383--1026 383--1113 383--1239 383--1529
## [64] 383--1648 383--1650 383--1651 383--1813 383--1864 385--501 385--558
## + ... omitted several edges
```

```
head(igraph::as_edgelist(nICTS_G10), 15)
```

```
##      [,1] [,2]
## [1,] "380" "383"
## [2,] "380" "385"
## [3,] "380" "388"
## [4,] "380" "394"
## [5,] "380" "396"
## [6,] "380" "404"
## [7,] "380" "464"
## [8,] "380" "470"
## [9,] "380" "472"
## [10,] "380" "478"
## [11,] "380" "501"
## [12,] "380" "549"
## [13,] "380" "550"
## [14,] "380" "571"
## [15,] "380" "618"
```

```
e <- igraph::as_data_frame(nICTS_G10, what="edges")
v <- igraph::as_data_frame(nICTS_G10, what="vertices")
```

```
kable(head(e, 15))
```

from	to
380	383
380	385
380	388
380	394
380	396
380	404
380	464
380	470
380	472
380	478
380	501
380	549
380	550
380	571
380	618

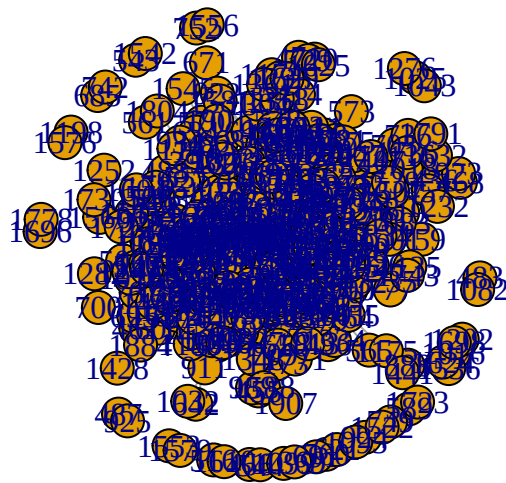
```
which(igraph::degree(nICTS_G10)==0)
```

```
## 415 449 469 493 525 535 587 599 649 670 707 716 724 730 747 751
## 18 22 35 47 57 60 87 91 119 127 147 151 156 160 169 171
## 763 765 878 894 928 933 950 951 953 1027 1034 1046 1077 1081 1096 1102
## 176 177 186 190 199 203 212 213 214 230 233 235 242 245 250 252
```

```
## 1110 1137 1176 1187 1228 1283 1284 1285 1286 1288 1296 1305 1318 1325 1345 1353
## 255 262 275 280 292 306 307 308 309 310 312 316 319 320 325 329
## 1357 1382 1395 1440 1467 1475 1492 1516 1540 1548 1554 1558 1562 1567 1568 1580
## 331 345 349 356 362 363 364 368 377 381 384 387 391 392 393 397
## 1582 1583 1587 1589 1590 1600 1643 1658 1659 1668 1675 1731 1739 1741 1742 1752
## 399 400 402 404 405 407 421 428 429 431 432 441 444 445 446 449
## 1773 1776 1795 1802 1806 1809 1818 1821 1823 1827 1865 1885 1891 1895 1904 1911
## 452 453 458 459 461 462 465 467 468 469 475 481 483 484 488 490

nICTS_G10_full <- igraph::delete.vertices(nICTS_G10, which(igraph::degree(nICTS_G10)==0))

set.seed(42)
plot(nICTS_G10_full, displaylabels=TRUE, boxed.labels=FALSE, label.cex=0.75)
```



Task 1: Paper and Network

Q1: Research question, context (what the study and network are about), main findings

Research question: The primary aim of this study is to explore how cross-disciplinary collaboration developed during ICTS funding at Washington University, using growth over time in scientific collaboration among ICTS members as a case. Specifically, the authors examine whether and how ICTS—the Institute of Clinical and Translational Sciences—enhanced cross-disciplinary collaboration in grant development and scientific publication over five years.

Context: With the rise of team science initiatives, a new field known as the science of team science has focused on evaluating and understanding how scientific collaborations form and develop. Traditional evaluation approaches tend to focus on separate stages of the research process—study planning and grant development, study implementation, and results dissemination and publication—but the authors argue that network analysis provides a more comprehensive and dynamic way to capture how interdisciplinary collaborations change over time.

This paper applies social network analysis (SNA) to the Washington University ICTS, which was established under the NIH Clinical and Translational Science Awards (CTSA) program in 2007. The analysis was conducted in two main ways: (1) Network mapping to visualize the growth of collaboration since ICTS inception. (2) Longitudinal analysis of five years of grant and publication data to evaluate the extent and evolution of cross-disciplinary collaboration.

Main findings:

- (1) Grant submissions: Under the Cohort Model, the number of ICTS members collaborating on grants increased, and indicators such as cross-disciplinary density ratio, E-I index, and modularity all showed

that a greater proportion of collaborations were cross-disciplinary in 2010 than in 2007. Under the Growth Model, collaboration also expanded, though the increase in cross-disciplinary ties was less pronounced, possibly because newer ICTS members were younger scientists facing funding pressures that limited cross-disciplinary collaboration. Overall, the growth in cross-disciplinary collaboration was more evident with the Cohort Model, which demonstrated a 61% decrease in modularity, compared with 49% in the Growth Model.

- (2) Publications: Under the Cohort Model, the size of the network and rate of collaboration increased, with more cross-disciplinary co-authorships in 2011 than in 2007. However, under the Growth Model, the cross-disciplinary density ratio and E–I index decreased, and modularity increased, indicating that newer members’ publications were less cross-disciplinary. The authors note that this may reflect the time lag between funding and publication and the career pressures on junior scientists to publish within their own disciplines.
- (3) Closer examination of interdisciplinary growth Monte Carlo simulations confirmed that the observed increases in collaboration were greater than expected by chance, especially for grants. Visual network maps showed that initially isolated groups—such as clinical or statistics clusters—became more connected and included a broader range of disciplines after three years of CTSA funding.
- (4) Conclusions and limitations: The study demonstrates clear growth in scientific collaboration among ICTS members over five years and an overall increase in cross-disciplinary collaboration. However, limitations include the lack of a non-ICTS comparison group, the single-discipline context, and the static definition of disciplinary identity. The authors emphasize that time remains a major barrier to achieving interdisciplinary work and that future research should develop better definitions and measures of disciplinarity.

Q2: Basic info about the networks, which you can obtain from R: number of vertices and edges (e.g., size); whether is directed or undirected; vertices and edges attributes; and any other details that you deem important

Number of vertices and edges The ICTS_G10 network represents collaboration ties among 493 members of Washington University’s Institute of Clinical and Translational Sciences (ICTS). It contains 1,359 edges, where each edge indicates that two scientists worked together on a grant submission in 2010. This serves as a concrete record of cross-disciplinary collaboration, since joint grant applications often bring together researchers from different fields.

Whether directed or undirected: The network is undirected, because collaboration on a grant is a mutual activity rather than a one-way relationship. vertices and edges attributes: Each vertex contains member-level information—year joined ICTS, discipline and an ID name. Edges are unweighted and represent whether a collaboration tie exists.

Additional details: The network is fairly sparse (density ≈ 0.011), meaning most researchers collaborated within limited circles. The network has a diameter of 9, suggesting that even in a large research community, information or resources can still flow through relatively few intermediaries.

Overall, the network structure reflects what Luke et al. (2015) describe as a “team science” environment — clusters of strong within-discipline collaboration linked by a few key cross-disciplinary connectors who help bridge different research areas.

Q3: Your assessment of the paper’s strengths and weaknesses. Include at least one aspect you particularly liked and why, and one aspect you did not like and why

Strengths

I really like the methodological clarity and innovation of this paper.

From the data selection and methodologies: The paper applied social network analysis to map and quantify patterns of collaboration over time. Instead of relying on self-reported surveys or administrative summaries, they used empirical network data from grant submissions and publications to trace how collaboration actually changed among ICTS members. This approach reflects a shift from qualitative description to

quantitative mapping in the evaluation of interdisciplinary research. The paper's use of cross-disciplinary density ratio, E-I index, and modularity provides a concrete, replicable framework to measure the extent of interdisciplinarity—something that many studies on team science only discuss in abstract terms.

From the perspective of the research context, the authors analyzed both grant development and publication networks, covering two essential stages of the research cycle. By comparing these two dimensions side by side, we can find that these two stages can both confirm and complement each other. So, it gives us a more complete picture of how collaboration emerges, evolves, and translates into outputs.

In terms of the visualization of collaboration networks. The figures showing the evolution from 2007 to 2010 make the findings tangible—we can literally see how disciplinary silos were being broken down. This visual storytelling helps communicate complex network concepts, making the study accessible to both researchers and practitioners who may not specialize in network analysis. In that sense, the paper not only contributes to academic theory but also provides a practical model for institutional evaluation.

Weakness:

Case-based and limited in scope. One major limitation of the paper is that it relies on a single institutional case study and provides little macro-level analysis. The focus is only on the Washington University ICTS, without a comparison group of non-ICTS members. As the authors themselves note, this design makes it difficult to determine whether the observed growth in cross-disciplinary collaboration is a direct outcome of ICTS funding or simply reflects a broader trend within the academic community.

Weak theoretical explanation. The analysis mainly depends on descriptive indicators such as density, E-I index, and modularity to show how collaboration changed, but provides little discussion of why these changes occurred or what mechanisms might explain them. Understanding interdisciplinarity requires not only measurement results but also an analysis of the underlying reasons. Some of the explanations offered—like time lag or funding pressures—remain surface-level and lack theoretical depth.

Insufficient support for the interpretation of publication data. I find the authors' explanation for the weaker evidence of cross-disciplinary co-authorship less convincing. They attribute it to "time lag" and "career pressures," but the paper provides no empirical or qualitative evidence to support this assumption. Including survey or interview data from ICTS members could have made this argument stronger and more convincing.

Static view of disciplinary identity. The study treats disciplinary identity as a fixed attribute, assigned once at the time of ICTS registration. In fact, scholars' disciplinary boundaries and collaborative roles are often dynamic and evolving—for example, a researcher might move from economics to sociology or combine expertise across different fields. This is especially true in long-term interdisciplinary projects. By overlooking this developing nature, the paper may underestimate the fluidity and complexity of collaboration networks.

No differentiation by field of collaboration. The paper does not distinguish between types of cross-disciplinary collaboration across different scientific fields. For instance, it highlights collaboration growth in health-related research, but what about the social sciences or arts? The extent and pattern of interdisciplinarity may differ substantially by field, yet we have no idea about it.

No attention to the degree of disciplinary distance. The study also fails to differentiate between collaborations across closely related disciplines and those that span distant fields. Empirically, interdisciplinary work often occurs between adjacent disciplines, while collaborations across distant areas are rarer and more challenging. It remains unclear whether the ICTS data support or contradict this tendency, as the paper does not address it.

Task 2: Centrality and Other Network Measures (Define and Compute)

1. Degree Centrality

Definition: The simplest definition of actor centrality is that central actors must be the most active in the sense that they have the most ties to other actors in the network or graph (Wasserman & Faust, 1994, p.178). Degree centrality can be defined as the number of ties incident upon a node (Borgatti, 2005, p.62).

Substantial definition: Number of ties a node has(Lecture 3 Slides, p.15). In social settings, it shows who interacts with many others and can reach people quickly. In Luke et al. (2015), a higher average degree meant researchers were collaborating more across disciplines over time, suggesting growth in cross-disciplinary ties and team science.

Calculation:

```
head(sort(igraph::degree(nICTS_G10_full), decreasing = TRUE), 15)
```

```
## 380 404 1409 947 383 472 531 645 644 1215 656 574 470 478 549
## 39 37 33 32 30 30 29 29 28 27 26 25 24 23 23
```

```
deg_n <- igraph::degree(nICTS_G10_full, normalized = TRUE)
head(sort(deg_n, decreasing = TRUE), 15)
```

```
##      380      404      1409      947      383      472      531
## 0.09848485 0.09343434 0.08333333 0.08080808 0.07575758 0.07575758 0.07323232
##      645      644      1215      656      574      470      478
## 0.07323232 0.07070707 0.06818182 0.06565657 0.06313131 0.06060606 0.05808081
##      549
## 0.05808081
```

2. Betweenness centrality

Definition: Betweenness centrality is defined as the share of times that a node i needs a node k (whose centrality is being measured) in order to reach a node j via the shortest path(Borgatti, 2005, p.60).

Substantial definition: Betweenness centrality shows how much a person connects different groups in the network. In Luke et al. (2015), people with high betweenness helped link researchers from different disciplines, making cross-team collaboration possible and speeding up information sharing.

Calculation:

```
head(sort(igraph::betweenness(nICTS_G10_full), decreasing = TRUE), 15)
```

```
##      380      549      645      404      383      644      472      394
## 10189.379 6965.198 5998.820 5716.347 5507.487 4248.690 4015.650 3878.236
##      726      703      1738      1409      656      745      1215
## 3827.759 3677.151 3370.402 3340.361 3275.317 2845.054 2799.623
```

```
btw_n <- igraph::betweenness(nICTS_G10_full, normalized = TRUE)
head(sort(btw_n, decreasing = TRUE), 15)
```

```
##      380      549      645      404      383      644      472
## 0.13028230 0.08905764 0.07670145 0.07308972 0.07041922 0.05432413 0.05134446
##      394      726      703      1738      1409      656      745
## 0.04958747 0.04894207 0.04701638 0.04309426 0.04271015 0.04187849 0.03637712
##      1215
## 0.03579623
```

3. Closeness Centrality

Definition: As defined by Freeman (1979), a node's closeness centrality is the sum of graph-theoretic distances from all other nodes, where the distance from a node to another is defined as the length (in links) of the shortest path from one to the other(Borgatti, 2005, p.59).

Substantial definition: Closeness centrality shows how quickly a person can reach everyone else in the network. In Luke et al. (2015), people with high closeness could share information and ideas faster across research teams, helping collaboration spread more efficiently through the network.

Calculation:

```
head(sort(igraph::closeness(nICTS_G10_full), decreasing = TRUE), 15)

## 466 483 487 516 543 544 589 685 742 749 752 924 925 1082 1095
## 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1

close_n <- igraph::closeness(nICTS_G10_full, normalized = TRUE)
head(sort(close_n, decreasing = TRUE), 15)

## 466 483 487 516 543 544 560 589 685 742 749 752 924 925 1016
## 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
```

4. Eigenvector Centrality

Definition: Eigen-vector centrality is defined as the principal eigenvector of the adjacency matrix defining the network(Borgatti, 2005, p.61). Eigenvector Centrality of a node measures its influence in terms of how connected the node is to other highly connected nodes(Lecture 3 Slides, p.40).

Substantial definition: Eigenvector centrality shows how well a person is connected to other well-connected people. In Luke et al. (2015), researchers with high eigenvector centrality were linked to key collaborators who themselves had many connections, giving them more influence and access to important parts of the research network.

Calculation:

```
head(sort(igraph::eigen_centrality(nICTS_G10_full, scale = FALSE)[[1]],
         decreasing = TRUE), 15)

## 404 470 478 1108 647 635 1215 640
## 1.0000000 0.8304045 0.8262700 0.7751209 0.7695187 0.7620336 0.7620168 0.7601446
## 678 630 639 719 733 1532 718
## 0.7553574 0.7471027 0.7427551 0.7388018 0.7355375 0.7304468 0.7304468

eig_n <- igraph::eigen_centrality(nICTS_G10_full, scale = TRUE)[[1]]
head(sort(eig_n, decreasing = TRUE), 15)

## 404 470 478 1108 647 635 1215 640
## 1.0000000 0.8304045 0.8262700 0.7751209 0.7695187 0.7620336 0.7620168 0.7601446
## 678 630 639 719 733 718 886
## 0.7553574 0.7471027 0.7427551 0.7388018 0.7355375 0.7304468 0.7304468
```

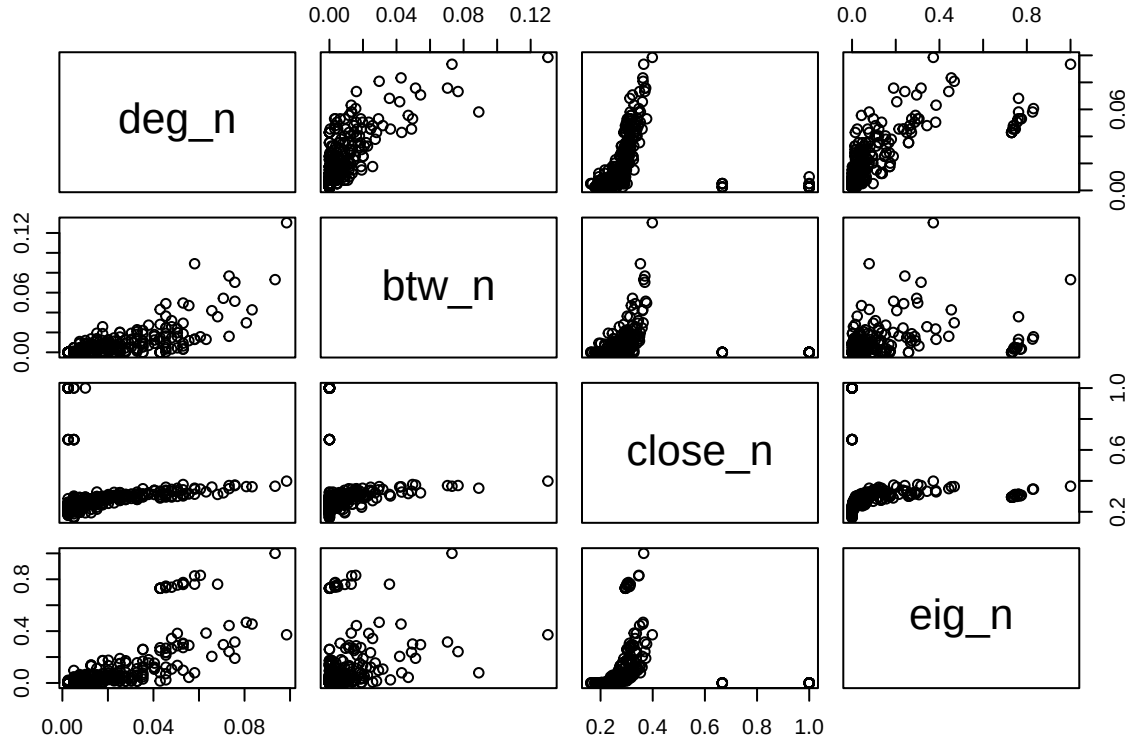
Comparison of Centrality Measures

```
all_centr <- data.frame(deg_n, btw_n, close_n, eig_n)
head(all_centr, 15)

## deg_n btw_n close_n eig_n
## 380 0.098484848 1.302823e-01 0.3984375 0.371908398
## 383 0.075757576 7.041922e-02 0.3695652 0.315978071
## 385 0.015151515 2.176206e-03 0.2914286 0.033135504
## 387 0.017676768 0.000000e+00 0.2921440 0.043752595
## 388 0.047979798 2.574194e-02 0.3327120 0.342911582
## 390 0.020202020 1.174104e-03 0.3005051 0.175144946
## 391 0.005050505 0.000000e+00 0.2611558 0.005139365
## 392 0.037878788 2.731373e-02 0.3230769 0.154301493
## 393 0.005050505 2.237566e-05 0.2075581 0.004039873
## 394 0.053030303 4.958747e-02 0.3765823 0.300579606
## 396 0.005050505 0.000000e+00 0.2921440 0.024145582
```

```
## 397 0.010101010 0.000000e+00 0.2613470 0.014958809
## 404 0.093434343 7.308972e-02 0.3654043 1.000000000
## 407 0.005050505 4.551848e-03 0.2260925 0.001429758
## 410 0.010101010 4.104400e-03 0.2590711 0.064939999
```

```
plot(all_centr)
```



```
cor(all_centr) %>% kable()
```

	deg_n	btw_n	close_n	eig_n
deg_n	1.0000000	0.7444206	-0.0964875	0.7354026
btw_n	0.7444206	1.0000000	-0.0247473	0.3352145
close_n	-0.0964875	-0.0247473	1.0000000	-0.0465982
eig_n	0.7354026	0.3352145	-0.0465982	1.0000000

Among the four measures, closeness centrality does not fit this network well. Closeness centrality assumes that the network is connected, so that the distance between any two nodes can be computed and interpreted (Wasserman & Faust, 1994, p. 183). However, our network is not fully connected, so the geodesic distances between some nodes are undefined or infinite, which violates the key assumption of full reachability required for closeness. As a result, closeness scores become unstable and show almost no correlation with other centrality measures, suggesting that this measure may not meaningfully capture influence or reach in this specific network.

5. Two Core Measures of choice

Density

Definition: Density is the proportion of observed ties (also called edges, arcs, or relations) in a network to the maximum number of possible ties. Thus, density is a ratio that can range from 0 to 1. The closer to 1 the density is, the more interconnected is the network (Luke, 2015, p.14).

Calculation:

```
igraph::graph.density(nICTS_G10_full)
```

```
## [1] 0.01728876
```

After moving out isolated nodes, the density is 0.017, showing that only a small fraction of possible collaborations actually occurred.

Diameter

Definition: The diameter then for an entire network is the longest of the shortest paths across all pairs of nodes (Luke, 2015, p.15).

Calculation:

```
igraph::diameter(nICTS_G10_full, directed = FALSE, unconnected = TRUE)
```

```
## [1] 9
```

The diameter is 9, meaning that even in a large institution, any two members can be connected through fewer than ten intermediaries.

Clustering Coefficient

Definition: The tendency to form closed triangles (Luke, 2015, p.16).

Calculation:

```
igraph::transitivity(nICTS_G10_full, type = "global")
```

```
## [1] 0.4833798
```

```
igraph::transitivity(nICTS_G10_full, type = "average")
```

```
## [1] 0.6683879
```

```
cmp <- igraph::components(nICTS_G10_full)
cmp$no
```

```
## [1] 18
```

The average clustering coefficient is high (over 0.6), suggesting that many collaboration groups form tight circles of three to five people. However, there's almost no cross-group collaboration, so the global network is relatively sparse and disconnected. Also, since `cmp$no` is quite large, it indicates the presence of many small components—meaning the network is composed of numerous high-density clusters.

6. Burt's Structural Holes: “Effective Size”, “Efficiency”, “Constraint”

Consider figure 1 of Borgatti's article “Unpacking Burt's Redundancy Measures”, reproduced below:

```
fig1 <- c(0,1,0,0,1,1,1,
          1,0,0,1,0,0,1,
          0,0,0,0,0,0,1,
          0,1,0,0,0,0,1,
          1,0,0,0,0,0,1,
          1,0,0,0,0,0,1,
          1,1,1,1,1,1,0)

fig1 <- matrix(fig1, nrow=7)

rownames(fig1) <- c(LETTERS[1:7])
colnames(fig1) <- c(LETTERS[1:7])
```

```
fig1
```

```
##   A B C D E F G
## A 0 1 0 0 1 1 1
## B 1 0 0 1 0 0 1
## C 0 0 0 0 0 0 1
## D 0 1 0 0 0 0 1
## E 1 0 0 0 0 0 1
## F 1 0 0 0 0 0 1
## G 1 1 1 1 1 1 0
```

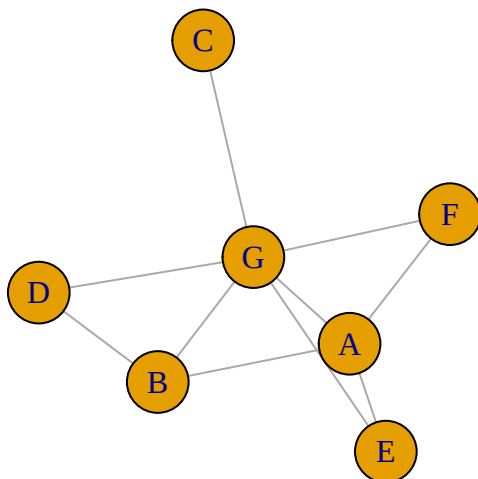
Using `fig1` above as input, calculate:

- “Effective Size” and “Efficiency”: calculate them by writing your own code in R (e.g., without using ready-to-use functions). Your code should make use of matrix algebra to generalize the in-class example (e.g., it should output a score for each node in the network). Briefly (1-2 sentences) interpret the results.
- “Constraint”: use `igraph::constraint()` to calculate it. Briefly (1-2 sentences) interpret the results.

Once you are sure the code produces the expected output, apply it to the network you are using for this assignment. Calculate all three measures on that network. Interpret the results in the context of your network. Note: the input should be a matrix (a square matrix of node–node ties).

```
graph <- graph_from_adjacency_matrix(fig1, mode = "undirected")
plot(graph, vertex.size = 30, main = "Figure 1 in Borgatti")
```

Figure 1 in Borgatti



```
n <- rowSums(fig1)
t <- rowSums( fig1 * (fig1 %*% fig1) )

eff_size <- n - (t / n)

data.frame(
  node = rownames(fig1),
  degree = n,
  t = t,
```

```
eff_size = eff_size,
row.names = NULL)
```

Effective Size

```
##   node degree t eff_size
## 1    A      4 6 2.500000
## 2    B      3 4 1.666667
## 3    C      1 0 1.000000
## 4    D      2 2 1.000000
## 5    E      2 2 1.000000
## 6    F      2 2 1.000000
## 7    G      6 8 4.666667
```

The results show that node G has the highest effective size (4.67), meaning it connects to others who are not strongly tied to each other, giving it access to more unique information. In contrast, nodes like C, D, E, and F have an effective size close to 1, meaning most of their contacts are connected with each other, so they get less new information from their ties and has less structural holes.

```
A_ICTS <- as.matrix(nICTS_G10_full, sparse = FALSE)

n_ICTS <- rowSums(A_ICTS)
t_ICTS <- rowSums( A_ICTS * (A_ICTS %*% A_ICTS) )

eff_size <- n_ICTS - (t_ICTS / n_ICTS)

head(data.frame(
  node      = V(nICTS_G10_full)$name,
  degree    = n_ICTS,
  t          = t_ICTS,
  eff_size  = eff_size,
  row.names = NULL
), 15)
```

```
##   node degree  t eff_size
## 1   380     39 250 32.589744
## 2   383     30 150 25.000000
## 3   385       6  16  3.333333
## 4   387       7  42  1.000000
## 5   388     19 104 13.526316
## 6   390       8  42  2.750000
## 7   391       2   2  1.000000
## 8   392     15 102  8.200000
## 9   393       2   0  2.000000
## 10  394     21 106 15.952381
## 11  396       2   2  1.000000
## 12  397       4  12  1.000000
## 13  404     37 408 25.972973
## 14  407       2   0  2.000000
## 15  410       4   6  2.500000
```

Nodes like 380 and 383 have large effective sizes, meaning they connect to many others who are not tightly linked with each other. This suggests they bridge across different disciplinary groups, similar to how Luke (2015) described researchers who connect separate academic fields and enable knowledge to move between disciplines.

```

n <- rowSums(fig1)
t <- rowSums( fig1 * (fig1 %*% fig1) )

eff_size <- n - (t / n)
efficiency <- eff_size / n

data.frame(
  node = rownames(fig1),
  degree = n,
  t = t,
  efficiency = efficiency,
  row.names = NULL)

```

Efficiency

```

##   node degree t efficiency
## 1    A      4 6  0.6250000
## 2    B      3 4  0.5555556
## 3    C      1 0  1.0000000
## 4    D      2 2  0.5000000
## 5    E      2 2  0.5000000
## 6    F      2 2  0.5000000
## 7    G      6 8  0.7777778

```

Nodes A, B, and G show higher efficiency, meaning they gain more unique information from their contacts. In contrast, nodes like D, E, and F have lower efficiency, indicating more redundant ties and overlapping information sources.

```

n_ICTS <- rowSums(A_ICTS)
t_ICTS <- rowSums( A_ICTS * (A_ICTS %*% A_ICTS) )

eff_size <- n_ICTS - (t_ICTS / n_ICTS)
efficiency <- eff_size / n_ICTS

head(data.frame(
  node      = V(nICTS_G10_full)$name,
  degree    = n_ICTS,
  t         = t_ICTS,
  efficiency = efficiency,
  row.names = NULL
), 15)

```

```

##   node degree  t efficiency
## 1   380    39 250  0.8356345
## 2   383    30 150  0.8333333
## 3   385     6  16  0.5555556
## 4   387     7  42  0.1428571
## 5   388    19 104  0.7119114
## 6   390     8  42  0.3437500
## 7   391     2   2  0.5000000
## 8   392    15 102  0.5466667
## 9   393     2   0  1.0000000
## 10  394    21 106  0.7596372
## 11  396     2   2  0.5000000
## 12  397     4  12  0.2500000

```

```
## 13 404      37 408 0.7019722
## 14 407       2  0 1.0000000
## 15 410       4  6 0.6250000
```

Nodes 380 and 383 have the highest efficiency (around 0.83), indicating that these researchers not only maintain many collaborations but also have little overlap among their collaborators, allowing them to efficiently access new knowledge across different disciplinary groups. This is consistent with Luke's (2015) findings, which suggest that such cross-disciplinary connectors play a key role in integrating ideas and speeding up knowledge exchange across different research fields

```
g <- graph_from_adjacency_matrix(fig1, mode = "undirected", diag = FALSE)
constraint <- igraph::constraint(g)

data.frame(
  node = V(g)$name,
  constraint = constraint,
  row.names = NULL)
```

Constraint

```
##   node constraint
## 1    A 0.5954861
## 2    B 0.6427469
## 3    C 1.0000000
## 4    D 0.7847222
## 5    E 0.7309028
## 6    F 0.7309028
## 7    G 0.4002701
```

Node G has the lowest constraint (0.40), meaning it connects otherwise separate nodes and acts as a bridge. In contrast, Node C has the highest constraint (1.00), showing it is fully tied to redundant partners within a closed circle.

```
constraint <- igraph::constraint(nICTS_G10_full)
head(data.frame(
  node = V(nICTS_G10_full)$name,
  constraint = constraint,
  row.names = NULL), 15)
```

```
##   node constraint
## 1   380 0.06132868
## 2   383 0.06922113
## 3   385 0.32712584
## 4   387 0.37886225
## 5   388 0.11037065
## 6   390 0.22560251
## 7   391 0.56215255
## 8   392 0.14834079
## 9   393 0.50000000
## 10  394 0.08928285
## 11  396 0.55638766
## 12  397 0.44633085
## 13  404 0.07891270
## 14  407 0.50000000
## 15  410 0.37814414
```


Nodes 380 and 383 show very low constraint values (around 0.06–0.07), suggesting they act as key bridges across disciplinary clusters. As Luke (2015) observed, such researchers often enable information to move across different fields, linking otherwise disconnected areas of team science.

Task 2: Centrality and Other Network Measures (Interpret)

Assessment of Measures

From the perspective of four centrality measures: The four centrality measures (degree, betweenness, closeness, and eigenvector), capture different aspects of influence and connectedness within the ICTS collaboration network. Degree and eigenvector centralities show a strong positive correlation ($r \approx 0.73$), indicating that individuals with numerous ties also tend to connect with other highly connected researchers. This pattern reflects the “hub” structure seen in Luke et al. (2015), where certain senior researchers act as central connectors in cross-disciplinary collaboration networks. Betweenness centrality, moderately correlated with degree ($r \approx 0.74$), identifies key brokers who link otherwise separated subfields—these researchers often help bridge disciplinary silos and accelerate information flow, similar to the bridging groups in Figure 3 of Luke’s paper.

Closeness centrality, however, performs poorly in this dataset. Since the ICTS network is not fully connected and contains many small components, the assumption of full reachability is violated. The resulting negative or near-zero correlations with other measures confirm that closeness is not meaningful here, it cannot reliably capture “how quickly” information travels across disconnected clusters. (More details can be found in the explanation under “Comparison of Centrality Measures”.)

Overall, centrality measures are useful for describing the formal structure of collaboration and identifying highly visible or influential researchers. Yet they mainly emphasize position rather than content: they show who connects to whom, but not what kind of knowledge is exchanged. Among them, betweenness centrality seems most informative for evaluating cross-disciplinary influence, because it identifies those who bridge separate research areas—an important feature in team science projects like ICTS.

In terms of Burt’s structural holes measures: Burt’s measures (effective size, efficiency, and constraint), offer a complementary perspective that focuses on the quality and redundancy of ties rather than their number. In the ICTS network, nodes 380 and 383 show very high effective size (≈ 32 and 25) and efficiency (≈ 0.83 – 0.84), along with extremely low constraint (≈ 0.06 – 0.07). This combination suggests that they connect otherwise separate clusters and access diverse, non-redundant information sources. In the words of Luke et al. (2015), such nodes bridge disciplinary communities and help integrate knowledge across biomedical, clinical, and social science fields.

Compared to traditional centrality, structural-hole measures better capture how unique or diverse one’s network connections are. Effective size and efficiency measure the extent of non-redundancy in one’s contacts, that is how many partners are not connected to each other, while constraint reflects the opposite, showing when ties are closed and overlapping. A low constraint value thus indicates strategic brokerage positions that are critical for interdisciplinary collaboration.

However, these measures also have limitations. They rely heavily on accuracy of local network structure; in sparse or fragmented networks, results can be sensitive to small changes in ties. Moreover, while they capture “opportunity for information brokerage,” they do not directly show whether the actor actually facilitates collaboration or knowledge transfer.

Overall, Burt’s measures are more conceptually aligned with the goals of team science, they emphasize bridging, diversity, and access to novel ideas. In this ICTS dataset, effective size and efficiency seem most insightful, as they clearly identify boundary-spanning researchers who connect clusters that would otherwise remain isolated.

Interpretation of Measures

Centrality Measures:

Firstly, high degree values in my results show that several researchers maintain many direct ties, forming

dense local circles. This pattern corresponds to what Luke et al. (2015) described as disciplinary hubs—groups that collaborate frequently within the same field. Such high-degree actors increase overall network density but do not necessarily foster cross-disciplinary integration.

Nodes 380 and 383 show exceptionally high betweenness, acting as bridging nodes that link separate clusters. This aligns with the “breaking down silos” process in the reading: these actors reduce modularity and increase E–I index, helping information move across disciplines (Luke, 2015). They represent the team science ideally, that is individuals who connect biomedical, clinical, and social science fields.

High eigenvector scores cluster around nodes with high degree, meaning influential actors are concentrated in core disciplines. This fits Luke et al.’s finding that early ICTS networks were dominated by senior clinical researchers. It represents reputation-based cohesion within existing fields rather than interdisciplinary expansion.

Closeness scores show no clear pattern because the network contains multiple disconnected components (as Luke’s early-year networks also did). Information cannot travel through shortest paths; instead, bridging via betweenness nodes drives communication.

Overall, the same dual pattern described in the readings applies here: Core cohesion within established disciplines (captured by degree/eigenvector). Interdisciplinary brokerage across clusters (captured by betweenness). However, the network still shows moderate modularity and many small components, meaning the cross-disciplinary process is developing but not yet dominant—just as Luke et al. observed between grant and publication phases.

Secondly, there are clear differences among the four centrality measures. As shown in my correlation matrix, degree and eigenvector are strongly correlated ($r \approx 0.73$), while betweenness only partly overlaps with them, and closeness shows almost no relationship. This means different centralities capture distinct social roles within the ICTS collaboration network.

The differences reflect two coexisting processes in ICTS: cohesion within disciplines is captured by both degree and eigenvector and bridging across disciplines is captured by betweenness. Because the network still shows moderate modularity and high clustering, measures emphasizing local density are more aligned, while those depending on global reach (e.g., closeness, betweenness) reveal only a few cross-disciplinary brokers.

Finally, the central social process in this network is the gradual shift from discipline-based collaboration toward cross-disciplinary integration. This reflects the process Luke et al. (2015) observed in the ICTS: the network expands from tightly knit, within-discipline clusters to more bridged and interconnected structures. In my results, the presence of multiple small but increasingly linked groups suggests that interdisciplinary brokerage is developing as a key dynamic.

In my view, betweenness centrality is best suited to describe this process. It reveals researchers who serve as boundary-spanners, connecting clusters that represent different disciplines or departments. In Luke et al.’s framework, these actors are essential for breaking down silos—they reduce modularity and increase the E–I index. My output shows exactly this: nodes with high betweenness also have low constraint and high efficiency, indicating that they occupy strategic bridging positions.

Structural Holes Measures:

Firstly, Burt’s structural holes measures show that the ICTS network remains highly clustered but only loosely connected across disciplines. The overall density (0.017) and the presence of 18 components from Assignment 1 confirm that collaboration is still fragmented into many small, discipline-based circles. Yet, nodes 380 and 383 stand out with very low constraint (≈ 0.06) and high effective size and efficiency, meaning they connect otherwise separated subgroups. These actors occupy brokerage positions that compensate for the network’s low density by creating cross-disciplinary ties—exactly the “breaking down silos” dynamic Luke et al. (2015) described.

This pattern aligns with the betweenness centrality results: high-betweenness nodes correspond to low-constraint brokers who link local clusters, while high degree and eigenvector scores mark the cohesive “disciplinary hubs.” Together, they reproduce the dual structure Luke observed—strong internal clustering but

increasing cross-field integration.

Secondly, Burt's structural-holes view assumes that people try to maximize access to non-redundant information and gain control advantages by positioning themselves between otherwise unconnected alters; maintaining many overlapping ties is costly and adds little value, whereas brokers benefit by spanning gaps across groups and disciplines. This logic fits the ICTS setting in the reading—team science grows when a few actors break down silos and connect communities. At the same time, it matches my network's structure from Assignment 1 (low density \$ 0.017, high average clustering \$ 0.66, many components), and results from Assignment 2 (nodes 380 and 383 have very low constraint \$ 0.06–0.07 and high effective size/efficiency), which is exactly what we would expect if they are boundary-spanners linking clinical, methods, and basic-science circles.

Other Core Measures:

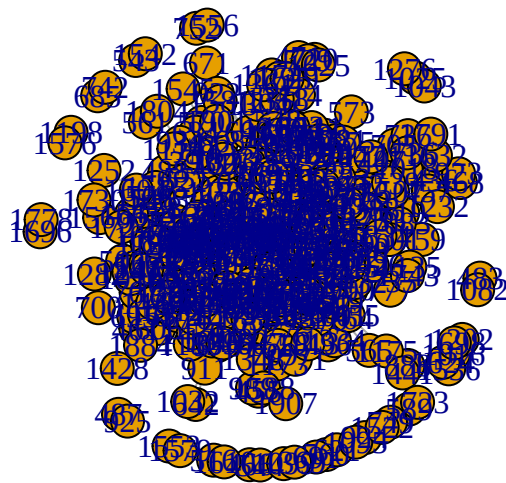
This week's measures (centrality and Burt's structural holes) connect directly with the earlier structural patterns. It still indicates a small world tendency at the local level (many dense triads of three to five collaborators), but it is not yet globally small-world, since the overall diameter of 9 and the presence of 18 components suggest limited connectivity across disciplines. Moreover, it further reveals who links those clusters and how information travels between them. High betweenness and low constraint identify a few brokers who create shortcuts across otherwise distant groups, functioning like the long ties that make small-worlds efficient. Meanwhile, degree and eigenvector still show the scale-free feature: most researchers have few ties, while a handful of hubs dominate the network. The presence of many closed triads explains why constraint is high for most actors; only a few with open triads enjoy structural holes and cross-disciplinary reach. In short, these new measures extend earlier theories: the ICTS network remains locally dense and globally sparse, but centrality and structural holes reveal the micro-mechanism: a few boundary-spanning actors connect otherwise separated clusters in this scale-free network, driving the growth of team science.

Task 3: Network Visualization

Default Visualization

```
set.seed(42)
plot(nICTS_G10_full, layout = layout_with_fr, main = "ICTS Network (Default)")
```

ICTS Network (Default)



The shortcomings of the default visualization: the default visualization looks cluttered and hard to read. Many nodes overlap, making it impossible to see individual labels or connections clearly. The dense cluster in the center hides structural patterns such as different groups or key connectors. Besides there is also no

color, size, or layout differentiation to help interpret the data — all nodes look the same, so important nodes or clusters cannot be distinguished.

Ggraph Visualization

```
nw_names <- network::get.vertex.attribute(ICTS_G10, "vertex.names")
disc_exp <- network::get.vertex.attribute(ICTS_G10, "DiscExp")
disc_col <- network::get.vertex.attribute(ICTS_G10, "DiscCol")
names(disc_exp) <- nw_names
names(disc_col) <- nw_names
```

```
g_names <- V(nICTS_G10_full)$name
V(nICTS_G10_full)$DiscExp <- disc_exp[g_names]
V(nICTS_G10_full)$DiscCol <- disc_col[g_names]
```

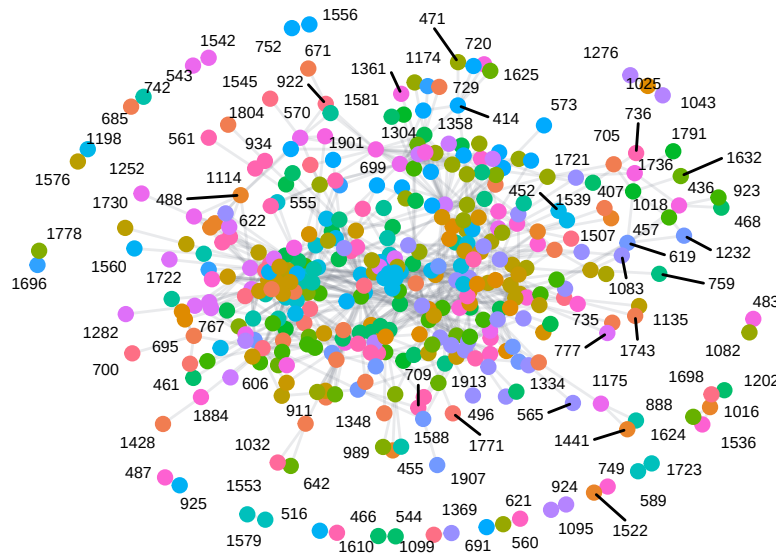
```
V(nICTS_G10_full)$label <- V(nICTS_G10_full)$name
```

```
set.seed(42)
nICTS_G10_full_viz <-
  ggraph(nICTS_G10_full, layout = "fr") +
  geom_edge_link(
    alpha = 0.15,
    colour = "#6b7280") +
  geom_node_point(
    aes(color = factor(DiscExp)),
    size = 2.2) +
  geom_node_text(
    aes(label = label),
    size = 2.2,
    repel = TRUE) +
  labs(
    title = "ICTS Collaboration Network",
    subtitle = paste("Nodes:", gorder(nICTS_G10_full),
                     "Edges:", gsize(nICTS_G10_full)),
    color = "Discipline"
  ) +
  theme_graph(base_family = "") +
  theme(legend.position = "right")
```

```
nICTS_G10_full_viz
```

ICTS Collaboration Network

Nodes: 397 Edges: 1359



Discipline

1100	6650
1111	6690
1200	6820
1600	6850
1700	6860
2000	6870
2200	6890
2400	6910
2800	6920
3300	6930
3600	6960
3900	6970
4200	6990
4400	7110
6100	7130
6500	7300
6520	7500

I replaced the default visualization with a Fruchterman–Reingold layout, colored nodes by DiscExp, and toned the edges way down (light, semi-transparent) so structure pops instead of noise. Nodes are a bit smaller and labels use repel so they don't sit on top of each other. I added a clear title, a concise subtitle with node/edge counts, and a right-side legend. The result is a cleaner map where clusters and bridges are visible and the plot is reproducible with a fixed seed.

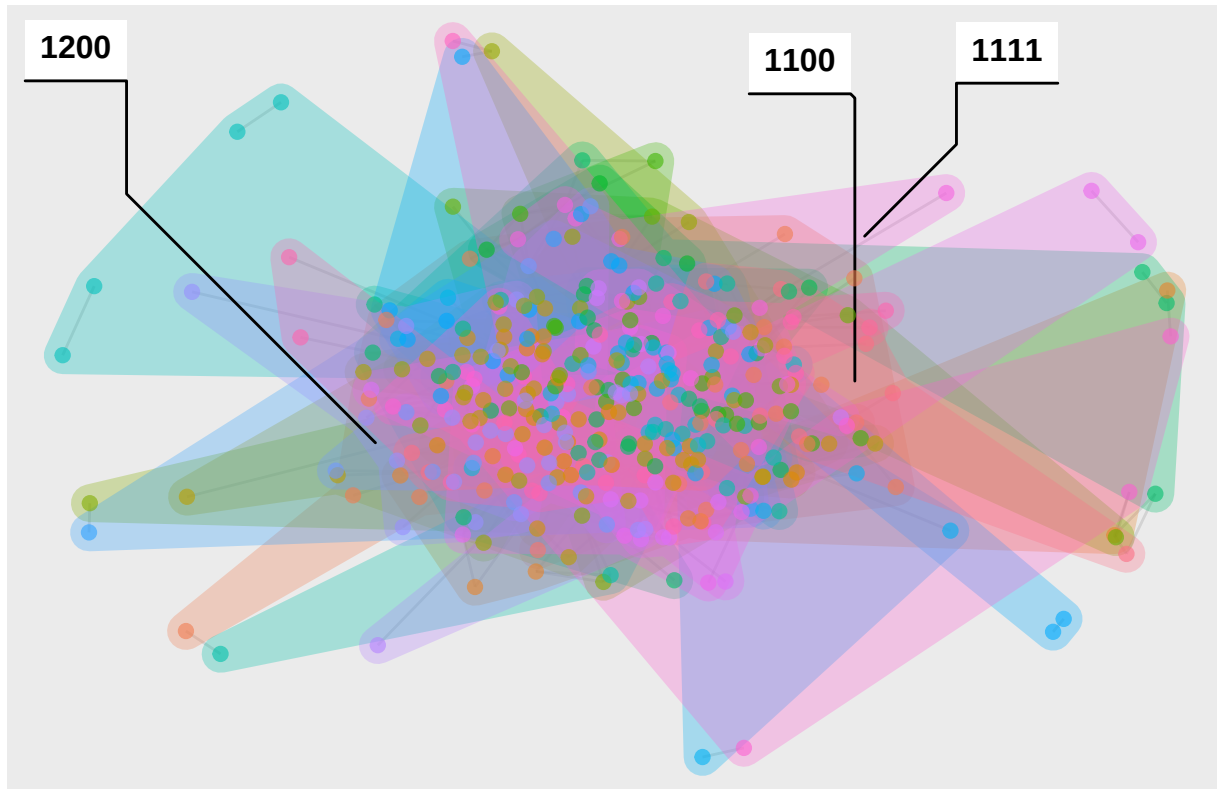
```
library(ggforce)
set.seed(42)

V(nICTS_G10_full)$DiscExp <- sub("-", "*", "", V(nICTS_G10_full)$DiscExp)

nICTS_G10_full %>%
  ggraph(layout = 'kk') +
    geom_edge_link(
      edge_alpha = 0.1,
      show.legend = FALSE) +
    geom_mark_hull(
      aes(x = x, y = y,
          group = factor(DiscExp),
          fill = factor(DiscExp),
          label = factor(DiscExp)),
      expand = unit(2.5, "mm"), alpha = 0.3,
      color = NA, show.legend = FALSE) +
    geom_node_point(
      aes(color = factor(DiscExp)),
      alpha = 0.6,
      size = 2.2,
      show.legend = FALSE) +
```

```
labs(title = "ICTS Collaboration Network")
```

ICTS Collaboration Network



Betweenness Centrality Visualization

```
V(nICTS_G10_full)$btw <- btw_n

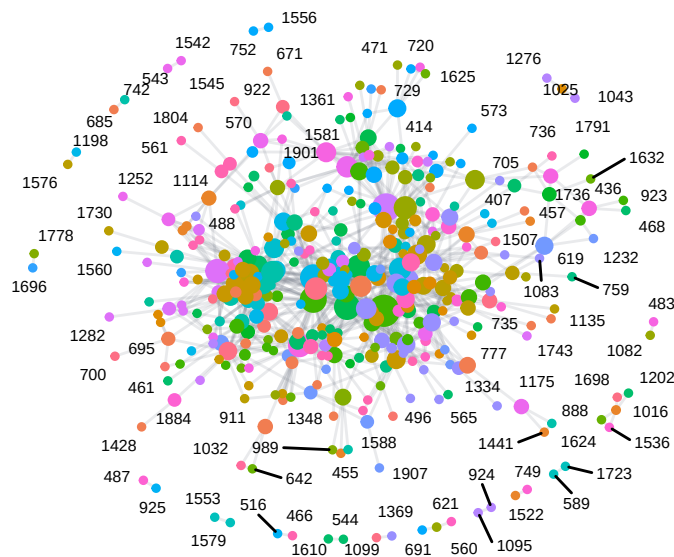
set.seed(42)
nICTS_G10_full_btw <-
  ggraph(nICTS_G10_full, layout = "fr") +
  geom_edge_link(
    alpha = 0.15,
    colour = "#6b7280") +
  geom_node_point(
    aes(color = factor(DiscExp),
      size = btw_n)
  ) +
  geom_node_text(
    aes(label = label),
    size = 2.1,
    repel = TRUE) +
  scale_size_continuous(
    range = c(1, 5),
    name = "Betweenness (normalized)") +
  labs(
    title = "Betweenness Centrality",
    subtitle = "Bigger nodes = stronger brokerage")
```

```
) +
theme_graph(base_family = "Helvetica") +
theme(legend.position = "right")

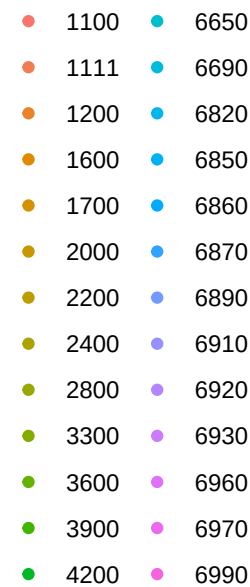
nICTS_G10_full_btw
```

Betweenness Centrality

Bigger nodes = stronger brokerage



factor(DiscExp)



Reflections & AI Use

Learning Reflections

I think the hardest part of this assignment was the operationalization and measurement. The ICTS network data is so huge, and there are too many measures to calculate. When writing the code, I had to modify it bit by bit, and even a small mistake would stop it from running. Another challenge was the interpretation — many parts felt repetitive, so for me it was difficult to explain them in a novel way.

AI Disclosure Statement

I used ChatGPT to help me revise and debug my code.

For example, I used `cmp <- igraph::components(nICTS_G10_full)` to identify the components;
`head(sort(btw_n, decreasing = TRUE), 15)` to show the top 15 most important nodes;

When plotting, I first aligned the vertex names with `nwn_names <- network::get.vertex.attribute(ICTS_G10, "vertex.names")`;

then created a `range = c(1, 5)` to separate node sizes and adjusted the visualization based on that.

References

Wasserman, S., & Faust, K. (1994). *Social network analysis: Methods and applications*. Cambridge University Press.

Luke, D. A., Carothers, B. J., Dhand, A., Bell, R. A., Moreland-Russell, S., Sarli, C. C., & Evanoff, B. A. (2015). Breaking Down Silos: Mapping Growth of Cross-Disciplinary Collaboration in a Translational Science Initiative. *Clinical and Translational Science*, 8(2), 143-149. <https://doi.org/10.1111/cts.12248>

Borgatti, S. P. (2005). Centrality and network flow. *Social Networks*, 27(1), 55–71. <https://doi.org/10.1016/j.socnet.2004.11.0>